Summary Notes: Photosynthesis

• Photosynthesis - Summary Notes (10 Lecture Slides)**

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SLIDE 1 - What Is Photosynthesis?

• Process that changes sunlight into chemical energy.

• Overall equation: **6 CO■ + 6 H■O + light -> C■H■■O■ + 6 O■**.

• Produces glucose (food) for the plant and oxygen for us.

• Basis of most food webs and the Earth's breathable air.

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SLIDE 2 - Where It Happens: The Chloroplast

• Tiny, double-membrane organelle in plant cells.

• **Thylakoids** - flattened sacs stacked into **grana**; site of light reactions.

• **Stroma** - fluid surrounding grana; site of the Calvin cycle.

• Pigments (chlorophyll a, chlorophyll b, carotenoids) sit in thylakoid membranes and give leaves their green colour.

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SLIDE 3 - Light-Dependent Reactions (Step 1)

• Take place in the thylakoid **membrane**.

• Sunlight excites pigments -> electrons are passed along a chain.

• Water is split (**photolysis**) -> releases O■, H■, and electrons.

• Energy from the electron flow pumps H■ into the thylakoid, creating a gradient that powers **ATP** production.

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SLIDE 4 - Photosystem II (PSII)

• Absorbs light at ~680 nm (red light).

• Gives electrons to the **electron transport chain**.

• Helps pump H■ into the thylakoid lumen.

• Output: electrons that will later be re-energised, plus O■ released to the atmosphere.

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SLIDE 5 - Photosystem I (PSI) & NADPH

• Absorbs light at ~700 nm (far-red light).

• Re-excites electrons coming from PSII.

• Final electron acceptor is **NADP■**, which becomes **NADPH** - a carrier of chemical energy.

• By the end of the light reactions we have **O■, ATP, and NADPH** ready for the next stage.

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SLIDE 6 - Calvin-Benson Cycle (Step 2)

• Occurs in the **stroma** using ATP & NADPH.

1. **Carbon fixation** - enzyme **Rubisco** attaches CO■ to RuBP -> makes 3-PGA.

2. **Reduction** - ATP + NADPH turn 3-PGA into **G3P** (a three-carbon sugar).

3. **Regeneration** - ATP helps rebuild RuBP so the cycle can continue.

• Some G3P leaves the cycle to become glucose and other sugars.

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SLIDE 7 - Rubisco & Photorespiration

• **Rubisco** = most abundant protein on Earth; key for fixing CO■.

• Works slowly and can mistakenly bind O■ instead of CO■.

• When O■ is bound, the plant wastes energy in a process called **photorespiration**.

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SLIDE 8 - C■ Plants (Typical Plants)

• Run the Calvin cycle directly in leaf cells (mesophyll).

• Efficient under cool, moist conditions.

• Lose efficiency when it's hot, dry, or CO■ is low (more photorespiration).

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SLIDE 9 - C■ and CAM Adaptations

• **C■ plants** (e.g., maize, sugarcane) - capture CO■ in mesophyll, then pass it to bundle-sheath cells where the Calvin cycle runs. This concentrates CO■ around Rubisco and cuts photorespiration.

• **CAM plants** (e.g., pineapple, many succulents) - open stomata at night, store CO■ as malic acid, and run the Calvin cycle during the day when stomata are closed, saving water.

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SLIDE 10 - Key Takeaways

• Photosynthesis turns light into glucose and O■ via two linked sets of reactions.

• Light-dependent reactions (thylakoids) make ATP, NADPH, and O■.

• Calvin-Benson cycle (stroma) uses those carriers to fix CO■ into sugars.

• Rubisco is essential but can cause wasteful photorespiration.

• C■, C■, and CAM pathways are plant strategies to keep photosynthesis efficient under different climates.

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• These ten slides give a clear, easy-to-read overview of photosynthesis for secondary-level students.*